

Polarized neutron diffraction with *ex-situ* ^3He neutron spin filter and two-dimensional detector for a complex incommensurate magnetic order

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Polarized neutron scattering has been widely used for studying magnetic orders in condensed matter. One of the typical setups for this technique is a triple-axis spectrometer with a ferromagnetic single-crystal monochromator and analyzer, which is suited for point-by-point measurements on a horizontal scattering plane. However, this setup cannot fully meet the demands of recent studies on complex magnetic orders with emergent cross-correlated phenomena. For instance, multiferroics, magnetic skyrmions, and other topologically-nontrivial magnetic orders tend to have incommensurate magnetic modulation vectors running along various directions in reciprocal space, which cannot be captured by measurements restricted to the horizontal scattering plane. To overcome these limitations, we developed an experimental method combining a ^3He neutron spin filter (^3He -NSF) with a two-dimensional (2D) detector, which enables polarization analysis of out-of-plane scattering over a wide Q region. This method was implemented on the polarized neutron triple-axis spectrometer PONTA at JRR-3 and applied to single-crystal diffraction measurements on the multiferroic material Mn_3WO_6 , which exhibits complex incommensurate magnetic orders. Polarization-dependent intensities of the incommensurate magnetic reflections were successfully observed. After correcting for the time-varying polarization of ^3He nuclear spins, these intensities were converted into three-dimensional distributions of spin-flip and non-spin-flip intensities in reciprocal space. These results demonstrate the effectiveness of the present method for analyzing complex magnetic structures.

I. INTRODUCTION

Polarized neutron scattering is a powerful technique for investigating magnetic structures and spin correlations because of the spin and associated magnetic moment of neutrons [1]. In particular, longitudinal polarization analysis, which analyzes the neutron spin states before and after scattering, enables the separation of magnetic and nuclear scattering, the separation of coherent and incoherent scattering, and the determination of magnetic moment directions [2]. In reactor-based neutron experiments, such polarization analysis is typically performed using the (111) plane of the ferromagnetic crystal Cu_2MnAl as the neutron polarizer and analyzer at triple-axis spectrometers [3–7], as shown in Fig. 1(a). While this type of instrument has been widely used for magnetic structure analysis, the accessible Q region for analyzing scattered neutrons is limited to the scattering plane. Therefore, it has been mainly applied to relatively simple magnetic structures.

In recent years, increasing attention has been paid to materials hosting complex magnetic structures. In such materials, the magnetic moments are often arranged in noncollinear or noncoplanar configurations, which can give rise to emergent electromagnetic responses and other

unconventional physical phenomena. Representative examples include nontrivial magnetic textures, such as magnetic skyrmions and spin-hedgehog lattices [8, 9]. A common feature of these systems is that their magnetic structures are often described by multiple magnetic modulation wavevectors, or multiple q -vectors. Consequently, the associated magnetic reflections can be distributed not only within a single horizontal scattering plane but throughout three-dimensional reciprocal space.

As an experimental technique for elucidating such complex magnetic structures, polarized small-angle neutron scattering (SANS) has been widely used [8, 10–12]. This is because, in polarized SANS, a large-solid-angle spin analyzer such as a polarized ^3He neutron spin filter (^3He -NSF) is typically used together with a two-dimensional detector to perform polarization analysis that includes the vertical scattering direction [13, 14]. In polarized SANS set up, the detector and the ^3He -NSF are positioned directly in line with the incident beam. This geometry is optimized for the measurement of relatively low- Q regions. However, polarization analysis must be extended to higher- Q regions to find numerous magnetic Bragg reflections to investigate complex magnetic structures more broadly. One promising approach is to combine a ^3He -NSF with a two-dimensional detector placed

at finite scattering angles, as illustrated in Fig. 1(b). Related instrumental concepts have already been reported at several neutron facilities, including POLI at MLZ [15] and WOMBAT at ANSTO [16]. Nevertheless, to the best of our knowledge, quantitative experimental demonstrations of polarization analysis using this type of setup remain limited.

In this work, we performed polarization analysis using a ^3He -NSF combined with a two-dimensional position-sensitive detector on the polarized neutron triple-axis spectrometer PONTA at JRR-3. Polarized neutron diffraction experiments on Mn_3WO_6 were performed using this setup. Mn_3WO_6 is a multiferroic material with a ferroaxial crystal structure belonging to the $R\bar{3}$ space group [17]. This system exhibits successive magnetic phase transitions at $T = 24.5$ K (antiferromagnetic transition) and 22 K, and spontaneous electric polarization has been reported to appear in the low-temperature magnetically ordered phase. These are demonstrated using polarization-analysis results obtained at 23 K, just below the antiferromagnetic transition. This paper mainly describes the experimental method and the data-analysis procedure and the details of the magnetic structure will be reported in a separate paper. Reciprocal-space mapping data are presented to demonstrate the usefulness of the present method for analyzing complex magnetic structures.

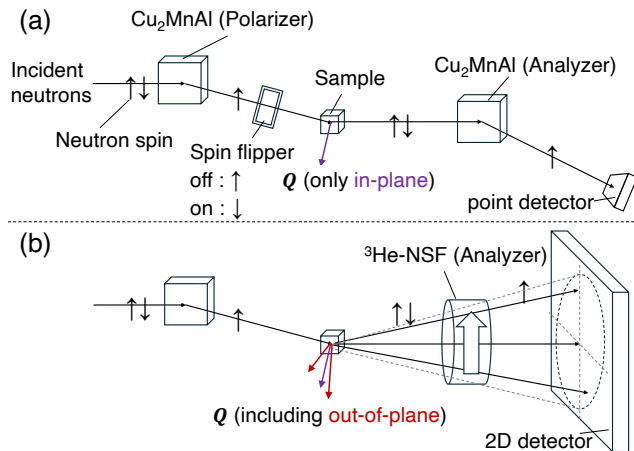


FIG. 1. Schematic illustrations comparing neutron polarization analysis techniques. (a) Conventional setup using a Cu_2MnAl crystal for both polarizer and analyzer, where only in-plane scattered neutrons are detected by a point detector. (b) Polarization analysis setup using a ^3He -NSF combined with a two-dimensional detector, enabling detection of both in-plane and out-of-plane scattered neutrons.

II. EXPERIMENTAL DETAILS

A. Setup

PONTA is one of the polarized neutron diffractometers at JRR-3, and it has a longitudinal polarization analysis option [7]. In this option, both incident and scattering neutrons are polarized and analyzed by Cu_2MnAl Heusler crystal monochromator. In this experiment, the Cu_2MnAl Heusler crystal analyzer and the point detector were not used; instead, the ^3He -NSF and a two-dimensional detector were installed. Figure 2 shows an overview of the experimental setup from the sample to the detector. The neutron detector was a scintillation-type two-dimensional detector with an active area of $256 \text{ mm} \times 256 \text{ mm}$ using wavelength-shifting fiber [18]. The detector and ^3He -NSF were positioned at an angle of approximately 32° relative to the incident neutron beam direction and fixed during the measurement. The sample-to-detector distance was approximately 0.9 m. To maximize the solid-angle coverage, the ^3He -NSF was placed as close to the sample as possible. The vertical angular coverage of the ^3He -NSF was $\pm 4.7^\circ$, enabling polarization analysis over approximately half of the detector area around the detector center. To suppress the background level, a B_4C -shielded neutron guide tube was installed between the ^3He -NSF and the detector. B_4C shielding was also wrapped around the ^3He -NSF coil. A single-crystal sample of Mn_3WO_6 with a volume of 8 mm^3 was mounted on an aluminum holder and installed in a cryostat with the scattering plane set to the (HOL) plane. The rotation angle ω of the sample was adjustable to align the target reflection with the detector. The Helmholtz coil was used to maintain the neutron spin polarization along the neutron flight path by applying a vertical magnetic field. The magnetic field in the ^3He -NSF was oriented parallel to the beam axis, and the spins of neutrons scattered from the sample adiabatically followed the field direction, as illustrated in Fig. 2. The magnetic field strength generated by both the ^3He -NSF coil and the Helmholtz coil was approximately 1 mT.

B. ^3He neutron spin filter

The neutron polarization principle of the ^3He -NSF is based on the strong spin dependence of the neutron absorption cross section of ^3He nuclei. Neutrons with spins antiparallel to the ^3He nuclear spins are strongly absorbed, while those with spins parallel to the ^3He nuclear spins are transmitted. Thus, neutrons are polarized by passing through the polarized ^3He gas. Among the methods for polarizing ^3He nuclei, spin-exchange optical pumping (SEOP) [19] and metastability-exchange optical pumping (MEOP) are the two major techniques. SEOP was adopted in this study because it has been continuously developed in Japan. In the SEOP method, a glass cell is filled with ^3He , N_2 , and a small amount of

TABLE I. Specifications of ^3He cells.

Name	$D \times L$ (mm)	Volume (cm^3)	^3He density (amg)	N_2 pressure (Torr)	K/Rb (mol)
Kenji	72×67	140	3.2	97	26
Shimpachi	72×65	146	3.1	100	37

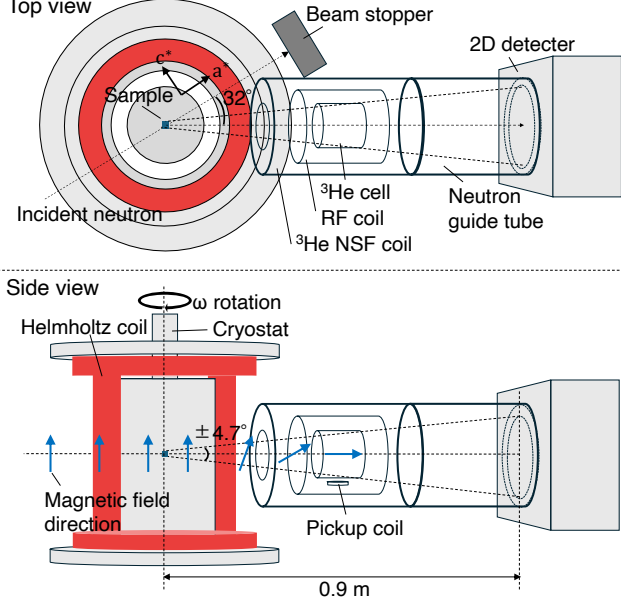


FIG. 2. Schematic illustration of the experimental setup at the 5G PONTA. Scattered neutrons from the sample pass through the ^3He -NSF coil and the neutron guide tube before reaching the two-dimensional detector. The sample is mounted in a cryostat that can be rotated by an angle ω . The magnetic field direction induced by the Helmholtz coil is vertical, whereas the field within the ^3He -NSF is oriented along the beam axis. The neutron spins are adiabatically rotated to follow the field direction. A radiofrequency (RF) coil is used to flip the ^3He polarization, and an NMR pickup coil monitors the ^3He polarization after each spin flip.

alkali metal, and is irradiated with circularly polarized light. The wavelength of the circularly polarized light corresponds to the transition wavelength of the alkali metal. The alkali metal electrons undergo repeated excitation and de-excitation, with N_2 serving as a quenching gas to suppress radiative decay, resulting in electron spin polarization. Then, the polarization of the alkali metal is subsequently transferred to the ^3He nuclei via spin-exchange collisions, thereby achieving ^3He nuclear polarization. The SEOP system used in this study was based on the system developed for the polarized neutron spectrometer POLANO at J-PARC MLF BL23 [20]. The SEOP system and the polarization analysis coil, shown in Fig. 2, are equipped with an adiabatic fast-passage nuclear magnetic resonance (AFP-NMR) system [21]. This system comprises a radio-frequency (RF) coil, which flips the ^3He polarization, and an NMR pickup coil placed

at the bottom of the ^3He cell, which monitors the ^3He polarization via the NMR signal. Two ^3He glass cells, named Kenji and Shimpachi, made of aluminosilicate glass (GE180) with nearly identical dimensions were used in this experiment. Both cells were filled with ^3He gas, nitrogen gas, and a mixture of alkali-metal atoms (Rb and K) for hybrid SEOP [22, 23]. The details of the cell specifications are described in Table I. A darkroom to house the SEOP laser system was constructed in the beamline experimental area, where the ^3He gas was polarized. After the ^3He polarization reached saturation, only the ^3He cell was transported to the ^3He -NSF coil on the beamline, where neutron measurements were performed during the relaxation of the ^3He polarization. This technique is known as the *ex-situ* ^3He -NSF method. Because the ^3He polarization decreases with time in this method, two cells were used alternately for the experiment. While one cell was used for the polarized neutron experiment, the other was polarized in the SEOP system, and the cells were exchanged every two or three days.

III. RESULTS

This section presents the results of polarized neutron diffraction experiments on a Mn_3WO_6 single crystal using a ^3He -NSF and a 2D detector. Figure 3 shows the procedure for data acquisition and reduction, which yields polarization-analysis results for magnetic reflections distributed in three-dimensional reciprocal space. A series of datasets was collected by scanning the sample rotation angle ω . At each ω , the ^3He polarization was flipped to measure both the non-spin-flip (I_{obs}^{++}) and spin-flip (I_{obs}^{+-}) scattering intensities. The observed scattering intensity, $I_{\text{obs}}^{\pm\pm}(t)$, is given by the following equation:

$$I_{\text{obs}}^{\pm\pm}(t) = N_0 \Sigma^{\pm\pm} T_p T_a(t) (1 \pm P_p P_a(t)), \quad (1)$$

where the superscripts of $I_{\text{obs}}^{\pm\pm}$ denote the neutron polarization states set by Cu_2MnAl and analyzed by the ^3He -NSF, respectively. N_0 is the number of incident neutrons from the reactor source. $\Sigma^{\pm\pm}$ represents the scattering cross section of the reflection of interest, and Σ^{++} and Σ^{+-} correspond to the non-spin-flip and spin-flip scattering cross sections, respectively. T_p is the neutron transmission of the Cu_2MnAl , $T_a(t)$ is the neutron transmission of the ^3He -NSF. P_p is the neutron polarization of the Cu_2MnAl , and $P_a(t)$ is the neutron polarization of the ^3He -NSF. The observed intensity $I_{\text{obs}}^{\pm\pm}$ is corrected for P_p , $P_a(t)$, and $T_a(t)$, yielding the corrected scattering

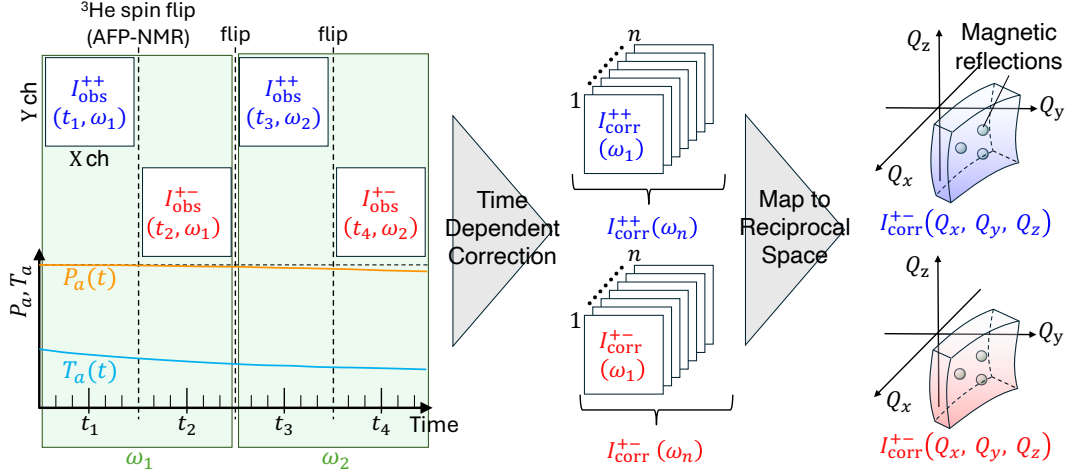


FIG. 3. Schematic diagram of the data-reduction procedure for polarization analysis using an *ex-situ* ^3He -NSF and a two-dimensional detector. Raw two-dimensional detector data measured at different sample rotation angles ω and times t are corrected using the time-dependent neutron polarization $P_n(t_i)$ and transmission $T_n(t_i)$. The detector coordinates (X,Y) and sample rotation angle ω are converted to Q_x, Q_y, Q_z , and then transformed to H, K, L using the UB matrix. The corrected three-dimensional reciprocal-space data can then be sliced on an arbitrary plane.

intensity $I_{\text{corr}}^{+\pm}$:

$$I_{\text{corr}}^{+\pm} = \frac{I_{\text{obs}}^{+\pm}}{T_a(t)(1 \pm P_p P_a(t))} = N_0 \Sigma^{+\pm} T_p. \quad (2)$$

In the *ex-situ* ^3He -NSF, the neutron polarization and transmission decrease with time owing to relaxation of the ^3He polarization. Therefore, it is necessary to evaluate the time dependence of the ^3He polarization. After correcting for the time-dependence of the polarization $P_n(t)$ and the transmission $T_n(t)$, the detector coordinates and sample rotation angle were then transformed into reciprocal-space coordinates. This transformation allows the polarization-analysis data to be obtained at arbitrary points in reciprocal space. The following subsections first describe how P_p , $P_a(t)$, and $T_a(t)$ were determined using the $\bar{1}02$ nuclear Bragg reflection of Mn_3WO_6 , and then present the corrected polarization-analysis results for the magnetic reflections.

A. Evaluation of ^3He neutron spin filter from nuclear scattering

^3He polarization is usually evaluated by measuring the transmission of a neutron beam through the ^3He cell. However, in this experiment, the ^3He cell cannot be placed in the direct path of the neutron beam because of the detector's sensitivity. Therefore, the ^3He polarization was evaluated from the nuclear Bragg peak intensity. The neutron transmission $T_a(t)$, the neutron polarization $P_a(t)$, and the ^3He polarization $P_{\text{He}}(t)$ are given by the

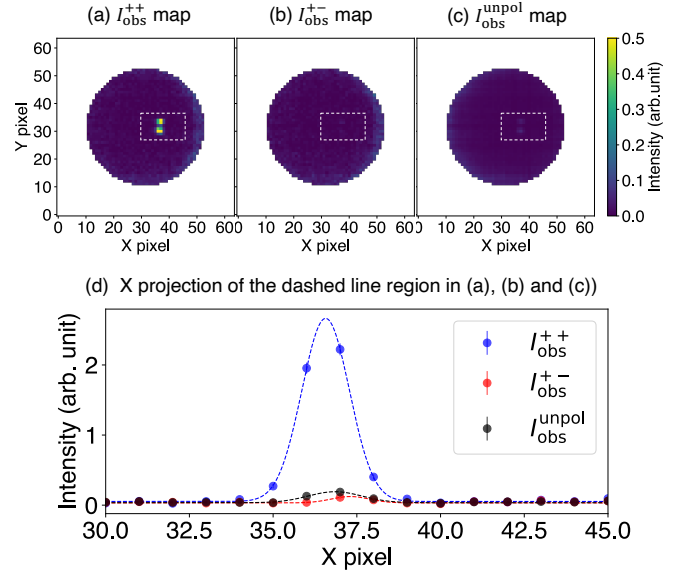


FIG. 4. (a) Observed non-spin-flip (I_{obs}^{++}), (b) spin-flip (I_{obs}^{+-}), and (c) unpolarized ($I_{\text{obs}}^{\text{unpol}}$) intensity maps, showing only the analyzing region. (d) Line profile along the x direction, obtained by integrating the intensity within the dashed regions in (a), (b), and (c).

following equations [24]:

$$T_a(t) = T_{\text{cell}} \exp(-\rho d \sigma(\lambda)) \cosh(\rho d \sigma(\lambda) P_{\text{He}}(t)), \quad (3)$$

$$P_a(t) = \tanh(\rho d \sigma(\lambda) P_{\text{He}}(t)), \quad (4)$$

$$P_{\text{He}}(t) = P_{\text{He},0} \exp(-t/\tau), \quad (5)$$

where T_{cell} is the transmission of the ^3He glass cell, ρ is the number density of ^3He , and d is the ^3He gas

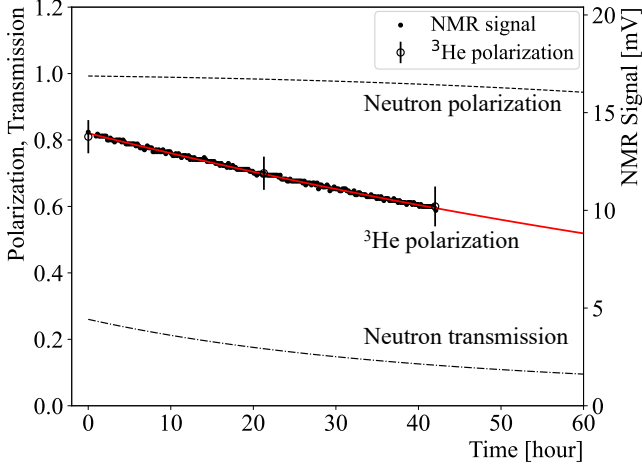


FIG. 5. Time dependence of the ^3He polarization, neutron polarization, and neutron transmission. The open circles show the ^3He polarization determined from the intensity of the $\bar{1}02$ nuclear reflection, while the closed circles show the ^3He NMR signal. The red line represents the fitting result using an exponential decay function. The dashed and dot-dashed lines indicate the calculated neutron polarization and transmission, respectively, obtained from the ^3He polarization.

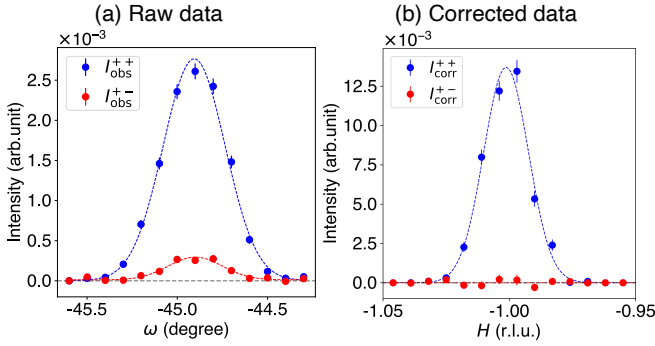


FIG. 6. The measured intensity of the $\bar{1}02$ reflection integrated over the peak region identified on the two-dimensional detector while scanning the sample rotation angle ω . (a) Raw data, where the red and blue symbols represent the non-spin-flip intensity I_{obs}^{++} and spin-flip intensity I_{obs}^{+-} , respectively. (b) Corrected data, where the red and blue symbols represent the corrected non-spin-flip and spin-flip scattering cross sections, Σ_{obs}^{++} and Σ_{obs}^{+-} , respectively.

thickness, $\sigma(\lambda)$ is the neutron absorption cross section of unpolarized ^3He , $P_{\text{He},0}$ is the initial polarization at $t = 0$, and τ is the relaxation time. The wavelength λ of the incident neutrons was monochromatized to 2.36 Å by Cu_2MnAl . By summing I_{obs}^{++} and I_{obs}^{+-} in Eq. 1, the following expression is obtained:

$$I_{\text{obs}}^{++}(t) + I_{\text{obs}}^{+-}(t) = 2N_0\Sigma^{++}T_pT_a(t). \quad (6)$$

Because the measurement time is much shorter than the relaxation time of the polarized ^3He gas, $I_{\text{obs}}^{++}(t)$ and

$I_{\text{obs}}^{+-}(t)$ can be regarded as being measured simultaneously. By taking the ratio with the intensity $I_{\text{obs}}^{\text{unpol}}$ measured when the ^3He polarization is zero ($P_{\text{He}} = 0$), the following equation is obtained:

$$\frac{I_{\text{obs}}^{++}(t) + I_{\text{obs}}^{+-}(t)}{I_{\text{obs}}^{\text{unpol}}} = 2 \cosh(\rho d \sigma(\lambda) P_{\text{He}}(t)). \quad (7)$$

The ^3He polarization $P_{\text{He}}(t)$ at the measurement time can be determined from these equations. The ^3He gas thickness ρd was determined from the transmission ratio between the ^3He cell and an empty cell of the same size. The value of ρd was 18.5 ± 0.5 [amg cm] for the Kenji cell and 18.8 ± 0.4 [amg cm] for the Shimpachi cell. Note that the amagat is a non-SI unit of the volumetric number density of an ideal gas at 0 °C (273.15 K) and 1 atm (101.325 kPa), and one amagat is equivalent to $2.69 \times 10^{19}/\text{cm}^3$.

Figure 4(a) and (b) show two-dimensional intensity maps of I_{obs}^{++} and I_{obs}^{+-} , respectively, measured just after transporting the polarized ^3He -saturated Shimpachi cell. Figure 4(c) shows the corresponding unpolarized intensity map, $I_{\text{obs}}^{\text{unpol}}$. Figure 4(d) shows the projection of the peak region of these intensity maps along the x axis. The ^3He polarization was estimated by summing the peak intensities of I_{obs}^{++} and I_{obs}^{+-} and taking the ratio with the peak intensity of $I_{\text{obs}}^{\text{unpol}}$. This yielded $P_{\text{He},0} = 0.81 \pm 0.05$ for the Shimpachi cell. Figure 5 shows the time evolution of the ^3He polarization determined from the nuclear reflection (open circles) and the NMR signal (closed circles). Throughout the experiment, the NMR signal was recorded during each magnetic scattering measurement at every ω angle. Nuclear scattering measurements were also performed periodically between the magnetic scattering measurements. The time evolution of the nuclear reflection intensities and the NMR signal showed good agreement. The relaxation time τ was estimated by fitting the NMR signal with Eq. 7, yielding $\tau = 131.1 \pm 0.7$ hours. The same measurement and analysis were also performed for the Kenji cell, yielding $P_{\text{He},0} = 0.72 \pm 0.06$ and $\tau = 128.0 \pm 0.3$ hours. Figure 5 also shows the neutron polarization and transmission calculated from Eq. 5 for the Shimpachi cell. These time-dependent values of the neutron polarization and transmission are the quantities required for the polarization correction.

The polarization of the Cu_2MnAl polarizer, P_p , was also determined from the same nuclear reflection data. From Eq. 1, the asymmetry between I_{obs}^{++} and I_{obs}^{+-} is given by

$$\frac{I_{\text{obs}}^{++} - I_{\text{obs}}^{+-}}{I_{\text{obs}}^{++} + I_{\text{obs}}^{+-}} = P_p P_a(t). \quad (8)$$

Because $P_a(t)$ is independently determined from the ^3He polarization, P_p can be obtained from this relation. The average value of P_p was found to be 0.86 ± 0.06 . Note that P_p includes the effects of environmental background and neutron depolarization in the guide field.

The validity of the polarization correction (Eq. 2) was confirmed using the ω scan of the $\bar{1}02$ reflection. At each ω , the intensity was integrated within the peak region identified on the two-dimensional detector (Fig. 4). The raw intensity profiles of the $\bar{1}02$ reflection show a finite spin-flip intensity owing to the imperfect neutron polarization, as shown in Fig. 6(a). By applying the polarization correction using $P_a(t)$, $T_a(t)$, and P_p , the corrected non-spin-flip and spin-flip intensities, I_{corr}^{++} and I_{corr}^{+-} , were obtained, as shown in Fig. 6(b). Since only the non-spin-flip scattering cross section Σ^{++} is expected for nuclear scattering, the result indicates that the polarization correction was successfully performed. Details of the polarization correction are described in Appendix A. These values were used for the polarization correction of the magnetic scattering data.

B. Polarization analysis of magnetic scattering

This subsection presents the polarization-analysis results for the magnetic reflections of Mn_3WO_6 at 23 K. First, intensity maps of antiferromagnetic reflections were measured using unpolarized neutrons and a two-dimensional detector. Because the scattering plane was set to the $(H0L)$ plane, a conventional triple-axis setup would mainly provide a reciprocal-space map in this plane, as shown in Fig. 7(d). However, the two-dimensional detector enables data analysis along the out-of-plane direction, as demonstrated in Fig. 7(b). Here, we can observe the three magnetic reflections $\mathbf{q}_1 = (0.059, 0.019, 0.5)$, and $\mathbf{q}_2$ and $\mathbf{q}_3$ correspond to three-fold rotations of $\mathbf{q}_1$ about the c^* axis. These $\mathbf{q}$ values show that $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ cannot lie within a single two-dimensional plane passing through the reciprocal-space origin (Fig. 7(a)). This highlights why three-dimensional reciprocal-space measurements are essential for capturing such magnetic structures. Figures 7(c) and 7(e) show the polarization-analysis maps obtained from the intensity difference $I_{\text{corr}}^{++} - I_{\text{corr}}^{+-}$. Clear positive and negative contrasts are observed at the magnetic reflection positions identified in the unpolarized intensity maps. In addition, to compare the spin-dependent intensities more quantitatively, line profiles along the H direction through $\mathbf{q}_3$ and $\mathbf{q}_1$ are shown in Figs. 7(f) and 7(g), respectively. The relative intensities of the non-spin-flip and spin-flip components differ clearly between the two reflections. These polarization-analysis data provide constraints on the magnetic moment components associated with each propagation vector. A detailed magnetic structure analysis based on these constraints will be presented in a separate paper. These results indicate that the spin-dependent scattering intensities were successfully resolved for magnetic reflections distributed in three-dimensional reciprocal space.

IV. SUMMARY AND OUTLOOK

In this work, we developed a polarization analysis method that combines ^3He -NSF with a two-dimensional detector to analyze arbitrary reciprocal-lattice points distributed in three-dimensional reciprocal space. We demonstrated a correction procedure for the time-dependent neutron polarization and transmission using a nuclear Bragg reflection and successfully verified its validity. Polarization analysis was successfully performed for the incommensurate magnetic reflections of Mn_3WO_6 at 23 K, and the details of the magnetic structure will be reported in a separate paper. The positions of the ^3He -NSF and the 2D detector were fixed in the present experiment, whereas a configuration in which both devices are mounted on a scanning arm is planned to cover a wider region of reciprocal space. The present *ex-situ* method requires this time-dependent correction. A compact *in-situ* ^3He -NSF system has recently been developed at J-PARC, in which SEOP is performed continuously during neutron scattering experiments to maintain stable neutron polarization [25]. This implementation is expected to maintain high neutron polarization and transmission throughout the experiment and thereby facilitate polarization analysis of complex magnetic structures in a broader range of materials.

ACKNOWLEDGMENTS

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Appendix A: Polarization correction with *ex-situ* ^3He neutron spin filter

In this section, the polarization correction for measurements using an *ex-situ* ^3He -NSF as an analyzer without a spin flipper is described. This correction is based on the polarization matrix method proposed by A. Wildes [26, 27]. The configuration of the apparatus used in this experiment is shown in Fig. 8. The spin state of the incident neutrons polarized by the Cu_2MnAl single-crystal monochromator is defined as (+). The spin state of the scattered neutrons analyzed by the ^3He -NSF is defined as (+) or (−), where the ^3He -NSF polarization can be flipped using AFP NMR. When the ^3He -NSF is polarized in the (+) state, the neutron intensity measured at the detector consists of four contributions, as illustrated

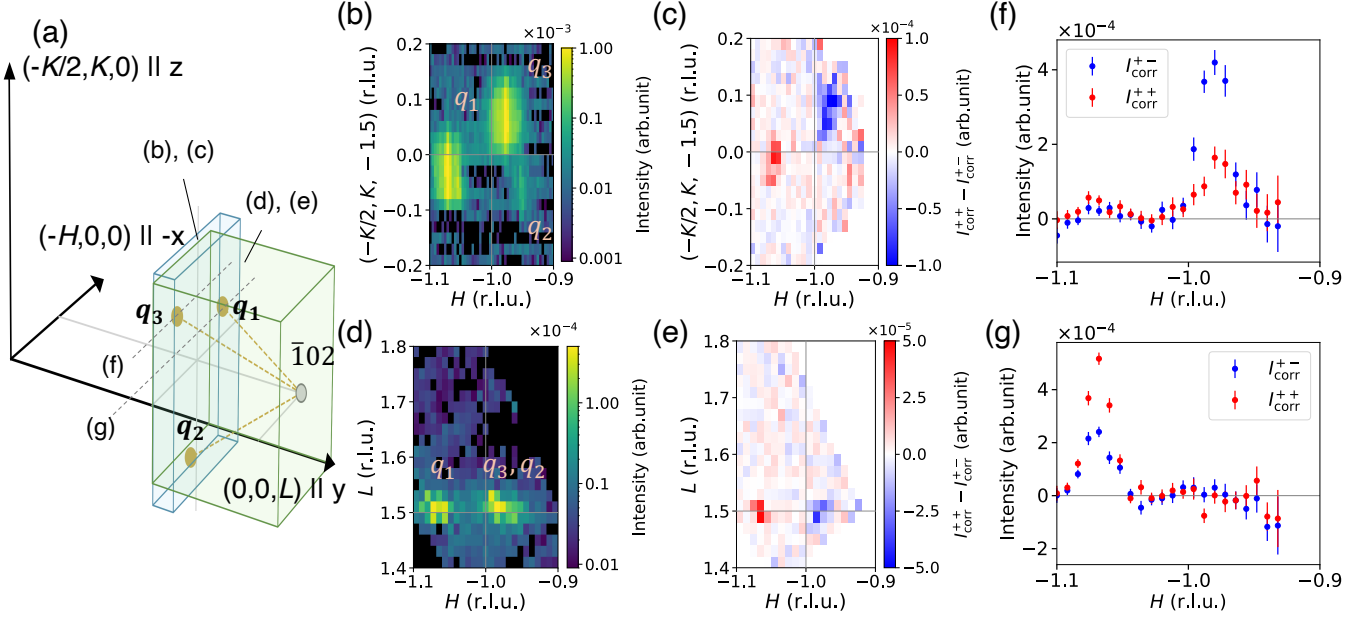


FIG. 7. (a) Schematic illustration of the reciprocal-space region measured in the present experiment. The translucent volumes labeled (b), (c) and (d), (e) indicate the regions integrated along the L direction and the $(-K/2, K, 0)$ direction, respectively. The dashed lines along the $-H$ direction through q_3 and q_1 indicate the line cuts shown in panels (f) and (g), respectively. (b), (d) Unpolarized neutron intensity maps in the $(H, K/2, -1.5)$ and $(H, 0, L)$ planes, respectively. (c), (e) Polarization-analysis maps obtained from the difference between the corrected non-spin-flip and corrected spin-flip intensities, $I_{\text{corr}}^{++} - I_{\text{corr}}^{+-}$. (f), (g) Line profiles of the (I_{corr}^{++}) and the (I_{corr}^{+-}) at q_3 and q_1 , respectively.

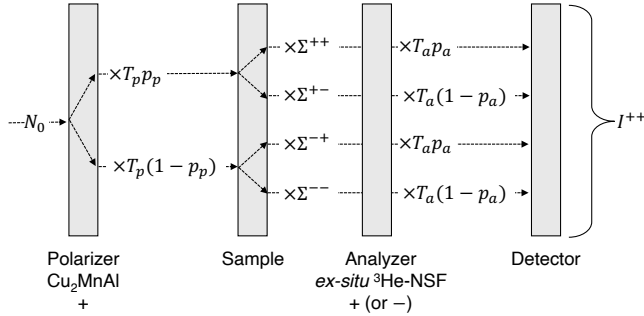


FIG. 8. Schematic diagram of the spin-dependent neutron intensity in the polarized neutron scattering experiment. The incident beam with intensity N_0 is polarized by the Cu_2MnAl polarizer and then scattered by the sample with the spin-dependent cross sections Σ^{++} , Σ^{+-} , Σ^{-+} , and Σ^{--} . The scattered neutrons are analyzed by the $^3\text{He-NSF}$ with transmission T_a and polarization efficiency p_a . The measured intensity I^{++} is given by the sum of the contributions from the four spin-scattering channels when the Cu_2MnAl and the $^3\text{He-NSF}$ are both polarized in the same direction.

in Fig. 8. It is expressed as

$$I^{++} = N_0 T_p p_p \Sigma^{++} T_a p_a + N_0 T_p p_p \Sigma^{+-} T_a (1 - p_a) + N_0 T_p (1 - p_p) \Sigma^{-+} T_a p_a + N_0 T_p (1 - p_p) \Sigma^{--} T_a (1 - p_a), \quad (\text{A1})$$

where N_0 is the number of incident neutrons, T_p and T_a are the total neutron transmissions of the polarizer and analyzer, p_p and p_a are the polarization efficiencies of the polarizer and analyzer, Σ^{++} , Σ^{+-} , Σ^{-+} , and Σ^{--} are the scattering cross sections for the corresponding spin states. The polarization efficiencies, p_p and p_a , are defined as $0.5 \leq p_p \leq 1$ and $0.5 \leq p_a \leq 1$, respectively. Their relation to the neutron polarization is given by $P_p = 2p_p - 1$ and $P_a = 2p_a - 1$. Only two configurations, I^{++} and I^{+-} , can be measured in this experimental setup. Assuming that the numbers of incident neutrons with spin (+) and (-) from the reactor neutron source are equal and that the scattering cross sections satisfy $\Sigma^{++} = \Sigma^{--}$ and $\Sigma^{+-} = \Sigma^{-+}$, Eq. A1 can be rewritten in the following matrix form:

$$\begin{bmatrix} I^{++} \\ I^{+-} \end{bmatrix} = \begin{bmatrix} T_p p_p T_a p_a + T_p (1 - p_p) T_a (1 - p_a) & T_p p_p T_a (1 - p_a) + T_p (1 - p_p) T_a p_a \\ T_p p_p T_a (1 - p_a) + T_p (1 - p_p) T_a p_a & T_p p_p T_a p_a + T_p (1 - p_p) T_a (1 - p_a) \end{bmatrix} \begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix}. \quad (\text{A2})$$

Since the *ex-situ* ^3He -NSF exhibits a decay of the ^3He polarization over time, both the neutron polarization efficiency p_a and the neutron transmission T_a become time

dependent. Accordingly, by treating Eq. A2 as a time-dependent matrix equation and taking its inverse, the following expression is obtained:

$$\begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix} = \frac{1}{T_p^2 T_a(t_1) T_a(t_2) P_p(P_a(t_1) + P_a(t_2))} \begin{bmatrix} T_p T_a(t_2)(1 + P_p P_a(t_2)) & -T_p T_a(t_1)(1 - P_p P_a(t_1)) \\ -T_p T_a(t_2)(1 - P_p P_a(t_2)) & T_p T_a(t_1)(1 + P_p P_a(t_1)) \end{bmatrix} \begin{bmatrix} I^{++}(t_1) \\ I^{+-}(t_2) \end{bmatrix}, \quad (\text{A3})$$

where p_p and p_a are transformed into P_p and P_a using the relation described above. The experimentally determined values of $T_a(t)$, P_p , and $P_a(t)$ are described in Sec. III A. While $T_a(t)$ and $P_a(t)$ vary with time owing to the relaxation of the ^3He polarization, T_p is constant

throughout the measurement. Therefore, T_p only gives a common scale factor to the scattering cross sections. Thus, this polarization correction yields the non-spin-flip and spin-flip scattering cross sections, Σ^{++} and Σ^{+-} .

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